



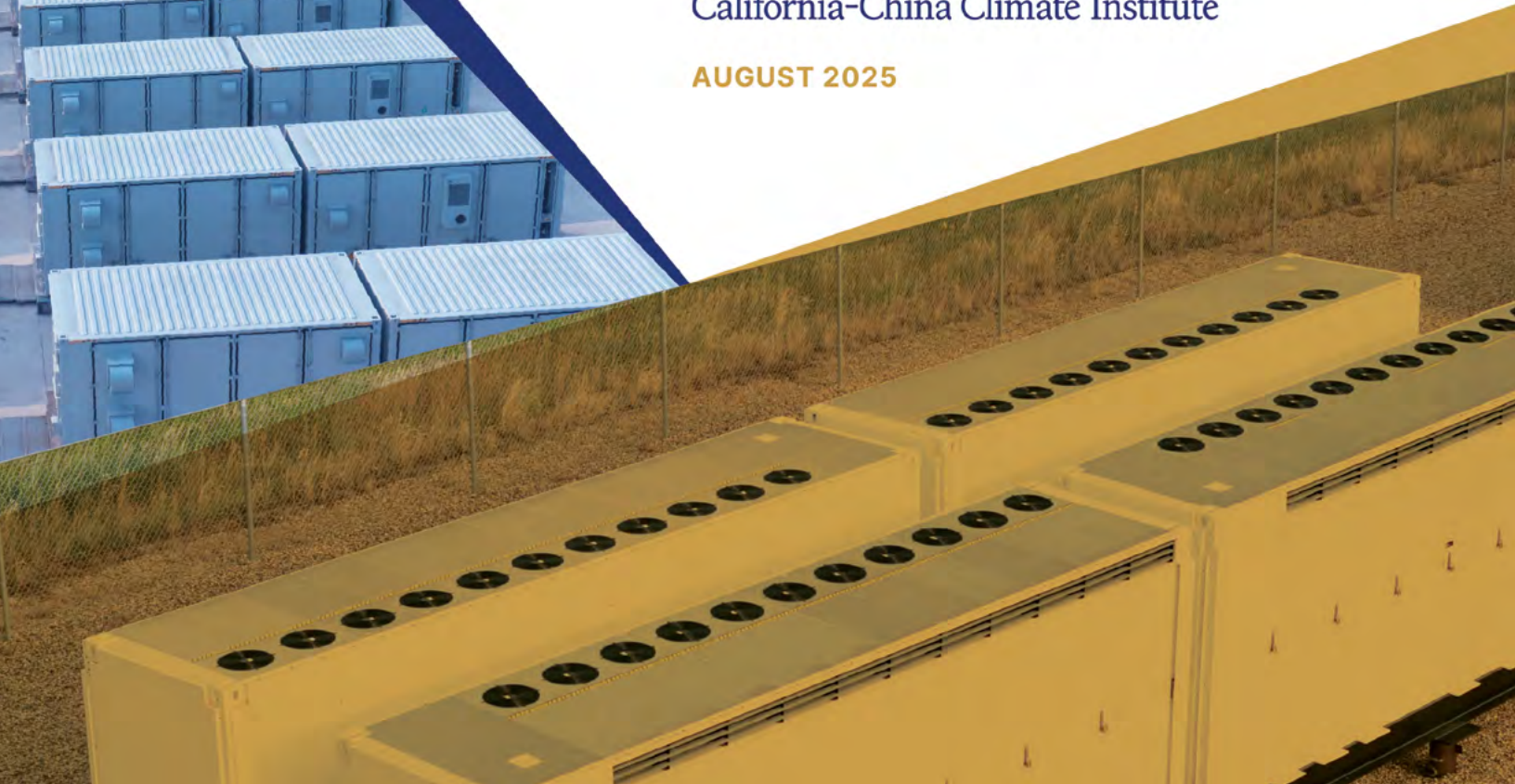
FINANCING BATTERY STORAGE IN CHINA:

INSIGHTS FROM CALIFORNIA'S PRACTICES



UC Berkeley
California-China Climate Institute

AUGUST 2025



AUTHORS

Anni Dai, Xueqin Cui, Jessica Gordon

ABOUT THE CALIFORNIA-CHINA CLIMATE INSTITUTE

The California-China Climate Institute The California-China Climate Institute was launched in September 2019 and is a university of Californiawide initiative housed jointly at UC Berkeley's School of Law and the Rausser College of Natural Resources. It is chaired by Jerry Brown, former Governor of the State of California. The Institute also works closely with other University of California campuses, departments and leaders. Through joint research, training and dialogue in and between California and China, this Institute aims to inform policymakers, foster cooperation and partnership and drive climate solutions at all levels.

ACKNOWLEDGEMENTS

This research has been supported by funding to the California-China Climate Institute at University of California, Berkeley. It has greatly benefited from the review and suggestions provided by Fredrich Kahrl, Max Dupuy (Regulatory Assistance Project) and Liu Jian (Energy Resources Institute).

EXECUTIVE SUMMARY

China's battery energy storage has expanded rapidly, driven by ambitious decarbonization goals and strong policy mandates. In 2021, the country set a national target of 30 GW of new energy storage capacity by 2025, a goal that was surpassed by the end of 2024, with total installed capacity reaching 73.76 GW. Despite this success in deployment, China still faces significant challenges in ensuring sustainable and market-based revenue streams for battery storage projects.

This report explores how California's experience in battery storage development can inform China's evolving market mechanisms. California's storage capacity grew tenfold between 2020 and 2023, supported by a comprehensive policy framework, multi-market participation opportunities, and robust capacity compensation mechanisms. Through comparative analysis, this report identifies key lessons that may help China design more effective and stable financing models for battery storage.

KEY TAKEAWAYS

- **California's multi-market approach offers strategic flexibility:** The state's Resource Adequacy (RA) program serves as a reliable framework for capacity payment through long-term bilateral contracts, while flexible participation in energy and ancillary markets allows batteries to capitalize on price spikes. Most large-scale battery projects are multi-market participants, underscoring the importance of diversified revenue streams and strategic flexibility. Among these markets, the capacity market is the most common form of participation.
- **Extreme weather events offer high-value opportunities:** Battery storage can capitalize on heatwaves and other grid stress events, reinforcing its role in enhancing grid reliability and generating outsized returns.
- **Stable income from capacity payments, with additional value captured in energy and ancillary service markets:** The Moss Landing energy storage project, the world's largest battery systems, secures 50-60% of its revenue through a RA capacity contract. Additional revenue gained via strategic bidding in energy and ancillary markets. Notably, 75% of its energy and ancillary market revenues were earned in just 15% of trading hours, emphasizing the value of flexibility and price responsiveness.
- **Toward a viable revenue model for battery storage:** The Moss Landing case illustrates a broader model combining stable capacity payments with market-based value capture. Two critical regulatory conditions are necessary to enable this model: (a) a competitive wholesale market that accurately reflects the value of flexible resources, and (b) a well-developed regulatory framework that supports long-term capacity payment.

POLICY RECOMMENDATIONS

- **Further developing spot markets** to unleash the market potential of battery storage and better reflect its true value as a flexible resource. Currently, no revenue models across Chinese provinces can capture the value that battery storage can offer.
- **Expanding ancillary services markets** to allow a broader range of services and adopt more market-reflective pricing structures.
- **Refining and scaling capacity compensation mechanisms**, encouraging Chinese provinces to move toward long-term capacity contracts for stable and predictable returns.
- **Establishing supportive regulatory frameworks**, including a more formal resource adequacy mechanism, to anchor long-term capacity payment for battery storage.

TABLE OF CONTENTS

1. Introduction	6
2. China's battery storage expansion and revenue models	7
2.1. Provincial revenue mechanisms and market models	8
2.2. Insufficient revenues: Varying compensation models in energy and ancillary service markets	10
2.3. Uncertain capacity payment: Short-term fixed rates and capacity leasing	11
3. California's battery storage: Regulatory framework and compensation mechanisms	13
4. Case study: Moss Landing energy storage project	16
4.1. Revenue streams	16
4.2. Operational pattern and compensation mechanisms	17
5. Conclusion and recommendations	21
References	22

1 INTRODUCTION

China's battery energy sector is experiencing rapid growth, driven by ambitious decarbonization goals and supportive policy mandates. In 2021, the country established a clear goal of reaching 30 GW of new energy storage capacity by 2025.¹ Remarkably, this goal was surpassed ahead of schedule, with installed battery storage capacity reaching 73.76 GW by the end of 2024,² more than doubling the original target. However, despite these impressive capacity additions, developing sustainable compensation mechanisms remains a key challenge. Battery storage projects across various provinces currently rely on different market models and compensation schemes, which often fail to provide stable and adequate revenues.

In this context, California's experience might offer valuable insights. California boasts one of the most dynamic battery storage markets in the U.S., with its installed capacity rapidly expanding from 500 MW in 2020 to 5,000 MW by May 2023.³ This growth has been supported by a comprehensive regulatory framework and robust compensation mechanisms, enabling rationally compensating revenue streams. These measures have not only facilitated substantial capacity expansion but also created a viable business case for storage investment.

This report examines how California's regulatory and market design can inform China's efforts to build sustainable, market-oriented compensation mechanisms for battery storage. By comparing provincial models in China with California's multi-market approach and diverse revenue streams, the report identifies strategies to enhance profitability and better integrate storage into China's evolving power market landscape.

The remainder of this report is structured as follows: Section 2 examines the development of battery storage in China, with a focus on emerging revenue models across provinces. Section 3 analyzes California's regulatory framework and outlines the primary revenue streams for battery storage projects over 100 MW. Section 4 presents a case study of the Moss Landing battery storage project, exploring its market participation strategies and studying how its value is captured in different markets. Section 5 concludes with policy implications and recommendations for a sustainable financing model for battery storage development in China.

¹ New battery storage refers to energy storage technologies other than conventional pumped hydro storage. These include advanced lithium-ion batteries, flow batteries, flywheels, compressed air energy storage, hydrogen (or ammonia) storage, and thermal energy storage. See Xinhua (2025a); NDRC and NEA (2021a).

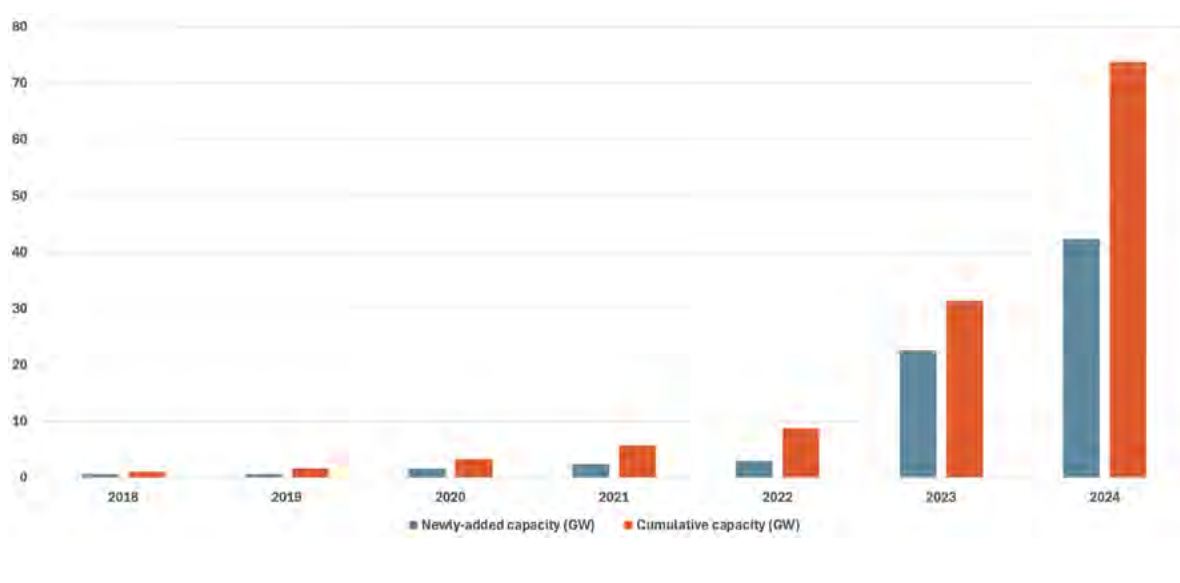
² Science and Technology Daily (2025).

³ California ISO (2023).

2 CHINA'S BATTERY STORAGE EXPANSION AND REVENUE MODELS

China's new energy storage capacity has expanded significantly, achieving a twenty-fold increase by the end of 2024 compared to the 2020 level (See Figure 1). This expansion has primarily been driven by lithium-ion batteries (See Figure 2). A key factor contributing to this growth was the “renewables plus storage” model, supported by a pivotal policy from the National Development Reform Commission (NDRC) and the National Energy Administration (NEA) in 2021.⁴ This policy required and incentivized the bundling of renewable energy generation with flexible resources, mainly battery storage,⁵ substantially accelerating the deployment of storage capacity.

FIGURE 1 Newly-installed and cumulative capacity of new energy storage in China

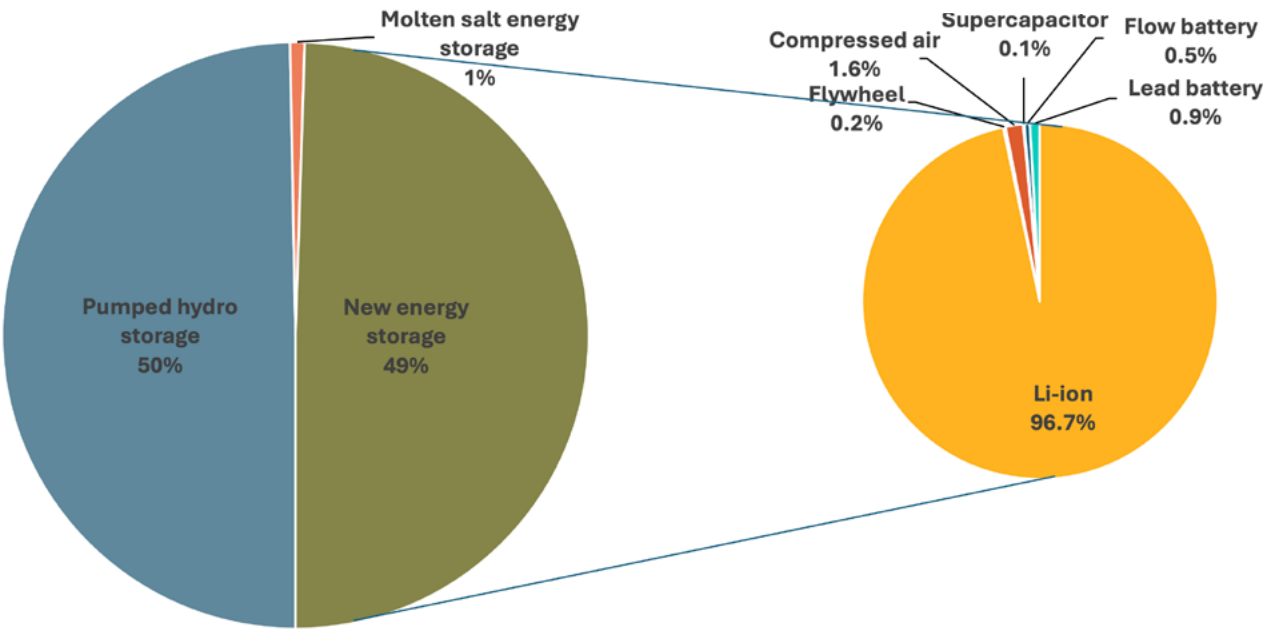


⁴ NDRC & NEA (2021b).

⁵ NDRC & NEA (2021b).

⁶ NEA (2025); NEA (2024); PwC China (2023); NEA (2023); BJX (2019); BJX (2020).

FIGURE 2 Distribution of cumulative energy storage capacity by technology (as of end - 2024)



2.1. PROVINCIAL REVENUE MECHANISMS AND MARKET MODELS

Chinese provinces are rapidly developing battery storage. Seventeen provinces had each reached over 1 GW of installed battery storage capacity by the end of 2024.⁸ In particular, following NDRC and NEA’s policy that allows battery storage to participate the power market,⁹ provinces have begun piloting various revenue mechanisms. Table 1 summarizes the revenue models employed in 12 provinces with over 2 GW of installed battery storage capacity, collectively accounting for 46.94 GW and 60% of the national total.¹⁰

TABLE 1 Business models across top provinces in grid-side storage capacity¹¹

	Energy market participation	Ancillary services	Capacity compensation
Inner Mongolia (West grid)	Participate spot market	Provides ancillary services	Pilot programs compensated at 0.35 RMB/ kWh for 10 years ¹²
Inner Mongolia (East grid)	No spot market; participate via peak, off-peak pricing or mid- and long-term agreement ¹³		

7 Sina Finance (2025).
8 BJX (2025).
9 NDRC & NEA (2023).
10 Top ten provinces are Xinjiang, Inner Mongolia, Shandong, Jiangsu, Ningxia, Gansu, Hunan, Hebei, Anhui, Guangxi, Zhejiang, Guangdong. See China Electricity Council (2025 March) BJX (2025).
11 Some of these provincial spot markets have yet to reach the full formal operational stage.
12 Xinhua (2025b).
13 Northeast Regulatory Bureau of the NEA, Inner Mongolia Department of Economy and Information Technology, Inner Mongolia Development and Reform Commission (2018).

	Energy market participation	Ancillary services	Capacity compensation
Xinjiang	No spot market; participates via peak, off-peak, and basic pricing with a 75% floating range ¹⁴	Charging during curtailment compensated at 0.55 RMB/kWh; discharging at 0.25 RMB/kWh ¹⁵	Capacity compensated at 0.2 RMB/kWh with a 20% annual reduction starting in 2024 ¹⁶
Shandong	> 5 MW with over 2 hour storage: eligible for day-ahead and real-time market ¹⁷	Various services paid at different prices, ¹⁸ including peak shedding, black start, ramp up/down	Support capacity leasing; Capacity compensated based on average volume traded during the peak periods ¹⁹
Jiangsu	>5 MW with over 2 hour storage: eligible for mid- and long-term agreement ²⁰	> 10MW with over 2 hour storage: eligible for load shedding and frequency regulation ²¹	Support capacity lease; ²² During January, July, August and December: compensated at 0.391 RMB/kWh with a decreasing rate. ²³
Ningxia	Participates in mid- and long-term transaction ²⁴	> 10 MW with over 2 hour storage allowed in providing load shedding ²⁵	Support capacity leasing for 4-hour storage ²⁶
Hebei	Participate in the spot market ²⁷	> 10 MW projects eligible in load shedding (south grid)	> 10 KW with over 2 hour storage compensated at 100 RMB/ kW per year, capped at 7.7 MW (south grid) and 8.3 MW (north grid) ²⁸
Zhejiang	Participate in day-ahead, real-time market, and mid- and long-term transactions	> 5 KW with over 2 hour storage eligible for providing load shedding services	For projects with > 600 utilization hours: annual compensation of RMB 200/MW (year 1), RMB 180/MW (year 2), RMB 170/MW (year 3) ²⁹
Gansu	Participate in day-ahead and real-time markets (15-minute price clearing) ³⁰	Provide load shedding service and frequency regulation ³¹	Compensated at RMB 300/MW per day; ³² Support capacity leasing

14 Xinjiang Development and Reform Commission (2023).

15 Xinjiang Development and Reform Commission (2024).

16 Xinhua (2025b).

17 RMI (2024); Shandong Energy Bureau. Shandong Province Implementation Rules for the Administration of Power Auxiliary Services (2023).

18 Shandong Energy Bureau. Shandong Province Implementation Rules for the Administration of Power Auxiliary Services (2023).

19 Shandong Energy Bureau. Shandong Province Implementation Rules for the Administration of Power Auxiliary Services (2023).

20 Jiangsu Development and Reform Commission (2023).

21 Jiangsu Regulatory Office of NEA (2024).

22 Jiangsu Development and Reform Commission (2023).

23 Jiangsu Development and Reform Commission (2023).

24 Ningxia Development and Reform Commission (2024a).

25 Ningxia Development and Reform Commission (2024b).

26 Ningxia Development and Reform Commission (2024b).

27 Hebei Power Trading Center (2024).

28 Hebei Development and Reform Commission (2025).

29 Zhejiang Development and Reform Commission & Zhejiang Energy Bureau (2021).

30 Gansu Department of Industry and Information Technology (2024).

31 Gansu Regulatory Office of NEA (2022).

32 Gansu Regulatory Office of NEA (2022).

	Energy market participation	Ancillary services	Capacity compensation
Guangdong	Participate in day-ahead, real-time, and spot markets ³³	Participate in the south regional ancillary market ³⁴	Compensated at 100 RMB/ MWh per year; ³⁵ Support capacity leasing ³⁶
Hunan	> 5MW storage can participate in spot market ³⁷ and mid- and long-term agreement ³⁸	Provide load shedding and frequency regulation ³⁹	Pilot storage capacity transaction under development
Anhui	Participate in day-ahead and real-time markets (5-minute price clearing) ⁴⁰	Participate in east regional market and compensated at 400 RMB/ MWh (frequency regulation) or 160 RMB/MWh (load shedding) ⁴¹	During January, July, August and December: Free charging; discharging compensated at 60% of on-grid coal-fired power prices and additional RMB 0.2/ kWh for traded energy ⁴²
Guangxi	No spot market; charging compensated at off-peak pricing, and discharging compensated at on-grid coal-fired power tariffs RMB 0.4207/kWh ⁴³	Participate in south regional grid market and compensated at RMB 396/ MWh (load shedding) ⁴⁴	Support capacity leasing ⁴⁵

2.2. INSUFFICIENT REVENUES: VARYING COMPENSATION MODELS IN ENERGY AND ANCILLARY SERVICE MARKETS

Drawing from Table 1, three dominant compensation models have emerged in energy and ancillary services markets across provinces, varying based on the different local market development summarized below:

	Revenues from energy markets	Revenues from ancillary services market
Model 1	Energy price from spot market	Providing a wider range of ancillary services
Model 2	Rates established through mid- and long-term contracts	Providing load shedding and frequency regulation
Model 3	Fixed peak/off-peak rates	Providing limited ancillary services compensated at fixed rates

³³ Guangdong Energy Bureau & Southern Regulatory Bureau of the NEA (2023).

³⁴ Guangdong Energy Bureau & Southern Regulatory Bureau of the NEA (2023).

³⁵ Guangdong Development and Reform Commission (2024).

³⁶ Guangdong Development and Reform Commission & Guangdong Energy Bureau (2023).

³⁷ Hunan Development and Reform Commission (2024).

³⁸ Hunan Regulatory Office of NEA (2025).

³⁹ Hunan Regulatory Office of NEA (2023).

⁴⁰ Anhui Power Trading Center (2024).

⁴¹ East China Energy Regulatory Bureau (2024).

⁴² Anhui Department of Industry and Information Technology (2024).

⁴³ Guangxi Development and Reform Commission (2023).

⁴⁴ South China Energy Regulatory Bureau (2025).

⁴⁵ Guangxi Development and Reform Commission (2023).

Model (1) is often adopted in provinces focusing more heavily on compensation from the spot energy market and ancillary service payments, such as Shandong, Guangdong, Jiangsu, Zhejiang, and Anhui. Battery storage projects in these regions actively participate in day-ahead and real-time market trading. Specifically, Shandong and Guangdong allow battery storage to provide broader ancillary services, including load shedding, frequency regulation, black start,⁴⁶ and ramping services.⁴⁷ Model (2) primarily relies on mid- and long-term agreements. Typically, battery storage systems above 5 MW or 10 MW qualify to provide load shedding and frequency regulation. Model (3) is used in provinces with less focus on compensation from the spot markets or ancillary service payments, such as Xinjiang, East Inner Mongolia, and Guangxi. These regions implement fixed peak/off-peak rates or on-grid coal-fired power benchmark tariffs to integrate storage resources. Ancillary services in these provinces are also compensated at fixed rates, generally between 0.25 RMB (\$0.035)/kWh and 0.55 RMB (\$0.076)/kWh.

Across these revenue models, delivering sufficient revenues remains a major challenge. For instance, Guangdong's 2024 spot market report highlights the difficulty storage projects face in achieving profitability in participating in energy and ancillary services markets. Stand-alone storage that participate in energy and ancillary services markets bid incurred an average charging price of 31.91 RMB (\$4.43)/MWh (totalling 0.5 billion kWh) and an average discharge price of 31.37 RMB (\$4.35)/MWh (totalling 0.44 billion kWh), resulting in a net loss of 21.38 million RMB.⁴⁸ Even if storage systems managed to leverage arbitrage, they would still face limited profitability due to minimal price volatility, the average price spread for storage was only 167 RMB (\$23.22)/MWh.⁴⁹

2.3. UNCERTAIN CAPACITY PAYMENT: SHORT-TERM FIXED RATES AND CAPACITY LEASING

Drawing from Table 1, eight out of twelve provinces offer short-term fixed-rate compensation mechanisms for battery capacity, typically 0.2-0.4 RMB (\$0.028-0.056)/kWh or 100-300 RMB(\$13.88-41.63)/KW annually. These rates usually extend over one to three years, with declining rates. A notable exception is West Inner Mongolia, which provides a 10-year fixed compensation at 0.35RMB (\$0.049)/kWh.

An emerging alternative revenue stream is capacity leasing, in which storage owners lease storage capacity, particularly to renewable energy developers who need to pair their projects with flexible resources. This model gained traction following the 2021 policy that established the “renewables plus storage” model. Since then, available battery capacity has increasingly become a prerequisite for renewable energy integration, and several provinces have issued binding requirements to ensure compliance with storage-to-renewables ratios. For instance, in Shandong, Guangdong, and Henan provinces, renewable projects that fail to meet storage portfolio requirements may be excluded from grid dispatch or procurement by utilities.⁵⁰

However, since 2025, China has begun unbundling battery storage from renewable resources, instead promoting its independent participation in power markets.⁵¹ This shift

⁴⁶ Black start refers to the ability of generation to restart parts of the power system to recover from a blackout. See NREL (n.d.).

⁴⁷ Ramping up/down refers to the ability of power plants or other resources to rapidly increase or decrease their power output, helping to maintain grid stability and balance supply and demand. See NREL (2020).

⁴⁸ Guangdong Power Exchange Center (2024).

⁴⁹ Guangdong Power Exchange Center (2024).

⁵⁰ China Energy News (2023); BJX (2023); General Office of Henan Government (2023); Guangdong Development and Reform Commission (2023).

⁵¹ NDRC& NEA (2025).

might remove a primary motivation behind the leasing model. Moreover, China's capacity market is yet to be developed, raising questions about appropriate pricing levels, duration of contracts, and stability of capacity compensation. These uncertainties might constrain long-term revenue predictability for battery storage investors.

Therefore, ongoing profitability issues in energy and ancillary service markets and the instability of current capacity compensation mechanisms may continue to hinder the financial viability and sustainability of battery storage in China, especially as batteries are encouraged to independently participate in power markets.⁵² In this context, insights gained from California might help illuminate how battery storage can participate in various market structures, how its value is captured, and how these lessons can inform the design of sustainable, market-based compensation mechanisms to support long-term battery storage investments in China.

⁵² NDRC & NEA (2025).

3 CALIFORNIA'S BATTERY STORAGE: REGULATORY FRAMEWORK AND COMPENSATION MECHANISMS

California power markets are regulated by two primary bodies: California Independent System Operator (CAISO) and California Public Utilities Commission (CPUC). Under the oversight of the Federal Energy Regulatory Commission (FERC), CAISO manages the wholesale electricity market,⁵³ while CPUC regulates investor-owned utilities (IOUs) and ensures reliable and affordable electricity for consumers.⁵⁴

In 2010, Assembly Bill 2514 - the state's first storage mandate - required CPUC to set procurement targets, resulting in a goal of 1,325 MW by 2020, distributed among three major IOUs:⁵⁵ Pacific Gas & Electric (580 MW), Southern California Edison (580 MW), and San Diego Gas & Electric (165 MW).⁵⁶ Subsequent laws further integrated storage into the state's renewable energy strategies. In 2015, Senate Bill (SB) 350 mandated 50% renewable electricity procurement by 2030 and directed CPUC and CAISO to integrate storage to address renewable intermittency.⁵⁷ SB 100, passed in 2018, raised the ambition further, requiring 100% zero-carbon electricity by 2045, thereby intensifying the demand for grid flexibility, particularly through battery storage.⁵⁸ At the federal level, FERC Order 841 required Regional Transmission Organizations and Independent System Operators to revise tariffs to explicitly facilitate energy storage participation in wholesale markets.⁵⁹

While federal and state legislative initiatives have provided the foundation for California's battery storage growth, CAISO developed market participation models that allow batteries to bid for charging and discharging in both day-ahead and real-time energy markets. Batteries have also become critical ancillary service providers, contributing to regulation up and down, frequency regulation, spinning and non-spinning reserves. Notably, batteries now supply more than half of CAISO's regulation up and regulation down requirements.⁶⁰

California does not operate a centralized capacity market. Instead, battery storage participates in the Resource Adequacy (RA) program, designed to ensure sufficient generation capacity for grid reliability and incentivize new resource development.⁶¹ Under the RA program, load-serving entities (LSEs) secure bilateral contracts with storage investors to meet their RA obligations.⁶² Since 2020, California has further refined the RA procurement, assigning three of the state's IOUs - PG&E, Southern California Edison,

⁵³ Federal Energy Regulatory Commission (n.d. -a).

⁵⁴ Six IOUs include Pacific Gas & Electric, Southern California Edison, San Diego Gas & Electric, Bear Valley Electric Service, Liberty Utilities and PacifiCorp. See Electric Investor Owned Utility Areas. (n.d.).

⁵⁵ CESA (2024).

⁵⁶ Utility Dive (n.d.).

⁵⁷ California Energy Commission (n.d.-a). Clean Energy and Pollution Reduction Act—SB 350. California Energy Commission. <https://www.energy.ca.gov/rules-and-regulations/energy-suppliers-reporting/clean-energy-and-pollution-reduction-act-sb-350>

⁵⁸ California Energy Commission (n.d.-b).

⁵⁹ Federal Energy Regulatory Commission (n.d.-b).

⁶⁰ California ISO (2023).

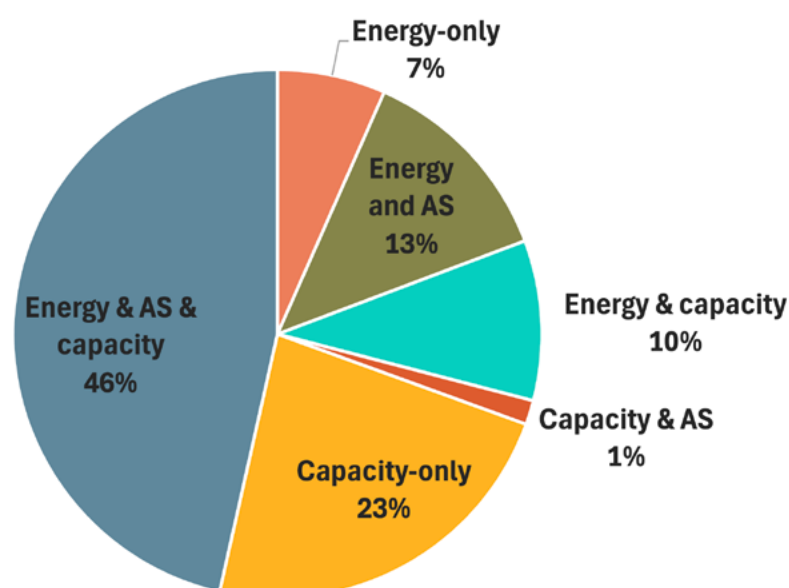
⁶¹ California ISO (2023).

⁶² McNamara, W. (n.d.).

and San Diego Gas & Electric - as central procurement entities responsible for securing local RA capacity on behalf of all LSEs.⁶³

Under California's regulatory structure, battery storage projects can access diverse revenue streams. Figure 3 examines the market participation patterns for battery storage projects exceeding 100 MW in size, which collectively represent 60% of the state's total installed capacity.⁶⁴ Most large-scale projects are multi-market participants; nearly half of their capacity simultaneously engage in energy, ancillary services, and capacity markets, highlighting a strong preference for diversified revenue streams and strategic flexibility. Moreover, the capacity market is the most common form of participation. Over 80% of battery storage capacity secures bilateral contracts, and nearly a quarter of capacity focusing exclusively on contracted capacity revenues. In contrast, energy-only strategy is relatively rare and less economically attractive.

FIGURE 3 Battery storage market participation distribution in California (Projects >100 MW)



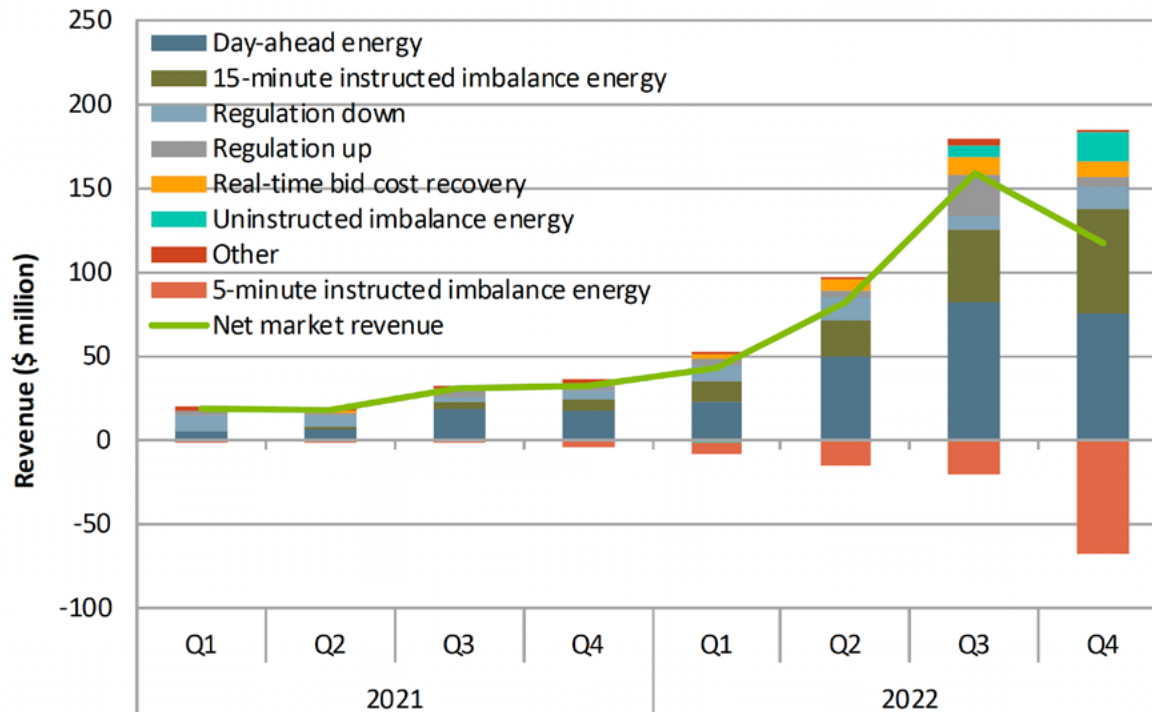
In the energy and ancillary services markets, day-ahead energy market schedules provide the largest and most consistent revenue stream, as demonstrated in Figure 4 from a CAISO special report. Figure 4 also indicates a substantial increase in the real-time market from 2021 to 2022, driven largely by battery storage systems responding flexibly to price spikes. Notably, an extreme heatwave in California during the summer of 2022 significantly boosted energy prices in Q3 and Q4. During this 10-day heat wave from August 31 to September 9 in 2022, battery storage projects generated nearly \$78 million, accounting for 20% of their total market earnings for the entire year.⁶⁵

⁶³ Resource adequacy requires three dimensional adequacy: (a) system resource adequacy, which requires load-serving entities to contract enough capacity to meet the 1-in-2 peak demand forecast, plus an additional planning reserve margin; (b) flexible resource adequacy, which requires load-serving entities to procure enough flexible capacity to meet the highest expected three-hour net load ramp each month; and (c) local resource adequacy, which requires load-serving entities to procure the capacity needed to meet demand in the event a local transmission grid emergency occurs. See California ISO (n.d.); McNamara, W. (n.d.).

⁶⁴ Electric Quarterly Reports (n.d.).

⁶⁵ California ISO (2023).

FIGURE 4 Quarterly revenues for battery (2021 -2022) by CAISO report



In summary, capacity payments serve as the most common revenue stream for battery storage projects in California. This is largely driven by the state’s RA program, which allows projects to secure long-term contracts for capacity and ensures grid reliability. Moreover, as most batteries are multi-market participants, the strategic flexibility of batteries is further captured in the energy and ancillary services market. In the competitive wholesale market, extreme weather like heat waves create price spikes and offer high-value opportunities for responsive and flexible resources. Battery storage projects are thus able to capitalize on these events, reinforcing their role in enhancing grid reliability and boosting revenues.

4 CASE STUDY: MOSS LANDING ENERGY STORAGE PROJECT

The Moss Landing energy storage facility in California stands as one of the world's largest four-hour lithium-ion battery energy storage systems. Owned and operated by Vistra Corp, an independent power producer, the facility has undergone multiple phases of development:

- **Unit 1:** commissioned in 2020, added 300 MW/1200 MWh
- **Unit 2 :** completed in 2021, contributed an additional 100 MW/400 MWh
- **Unit 3 :** finalized in mid-2023, added 350 MW/1400 MWh

Together, these expansions bring the facility's total capacity to 750 MW/3000 MWh.

4.1. REVENUE STREAMS

For the Moss Landing storage project, the capacity market provides the most stable and significant source of revenue across all units (see Figure 4). This facility benefits from a 20-year RA procurement agreement with PG&E. Figure 4 illustrates that across all three units, capacity payments make up between 50% and 61% of total revenues, paid at \$7600/MW per month.

This long-term RA agreement supports PG&E's 580 MW procurement target, as required by CPUC, and is primarily driven by local reliability needs. In particular, the retirement of dispatchable generation assets in the South Bay - Moss Landing subarea created a localized capacity shortfall, prompting the solicitation of battery storage to ensure grid reliability.⁶⁷ The associated costs are thus recovered through California's Cost Allocation Mechanism (CAM), which spreads the expense across all electricity customers in the service territory via a non-bypassable charge, regardless of their energy provider.⁶⁸

In addition to contracted capacity revenues, Moss Landing storage units also bid into the energy and ancillary service markets. However, these revenues are more volatile. Unit 1 reported net energy market losses of -\$107,762 in Q4 2021 and -\$16,194 in Q1 2022. In contrast, Unit 1 and Unit 2 both saw significant revenue increases in Q3 2022 driven by extreme summer heatwaves, during which batteries capitalized on elevated market prices.

⁶⁷ California Public Utilities Commission (2018)

⁶⁸ California Public Utilities Commission (2018)

4.2. OPERATIONAL PATTERN AND COMPENSATION MECHANISMS

As a highly flexible resource, Moss Landing energy storage plays an important role in addressing solar intermittency and responding to real-time price signals in the regional grid. Figure 5 (a) represents average hourly charging (negative quantity) and discharging (positive quantity), along with corresponding transaction charges (earnings or costs) by hour as shown in Figure 5 (b), aggregated on a monthly basis. The data in Figure 5(a)

FIGURE 5 Quarterly revenue breakdown for Moss Landing Energy storage across units



shows that on a given day, charging (buys energy) is concentrated during daytime hours when regional solar generation peaks, while discharging (sells energy) is heavily clustered in the late afternoon and evening hours when demand and prices rise.

Moss Landing storage project also effectively responds to monthly price variations to optimize arbitrage. As Figure 6 illustrates, in July, the project engaged in deep charging (up to -200 MWh) followed by high-value discharges exceeding \$ 30,000 per hour. This trend is confirmed in Figure 7, which shows average hourly energy arbitrage for each quarter in 2024. The aggressive charging/discharging is driven by larger arbitrage in summer time. In comparison, the hourly average arbitrage spreads in California reached \$75/MWh during Q3 2024, substantially higher than \$22/MWh average in Guangdong's 2024 spot market.

FIGURE 6 (a) Hourly average transaction quantities in 2024 by month

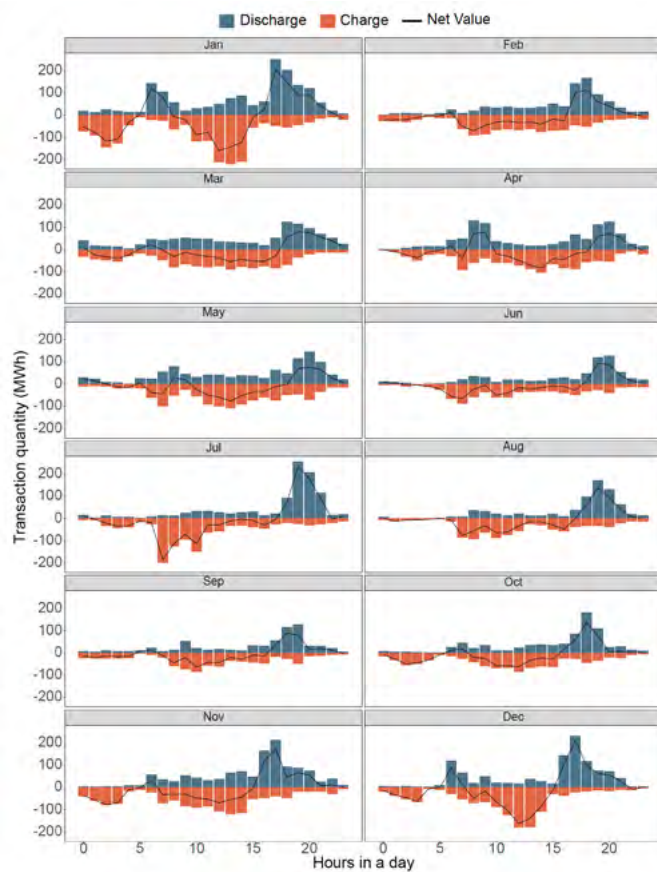
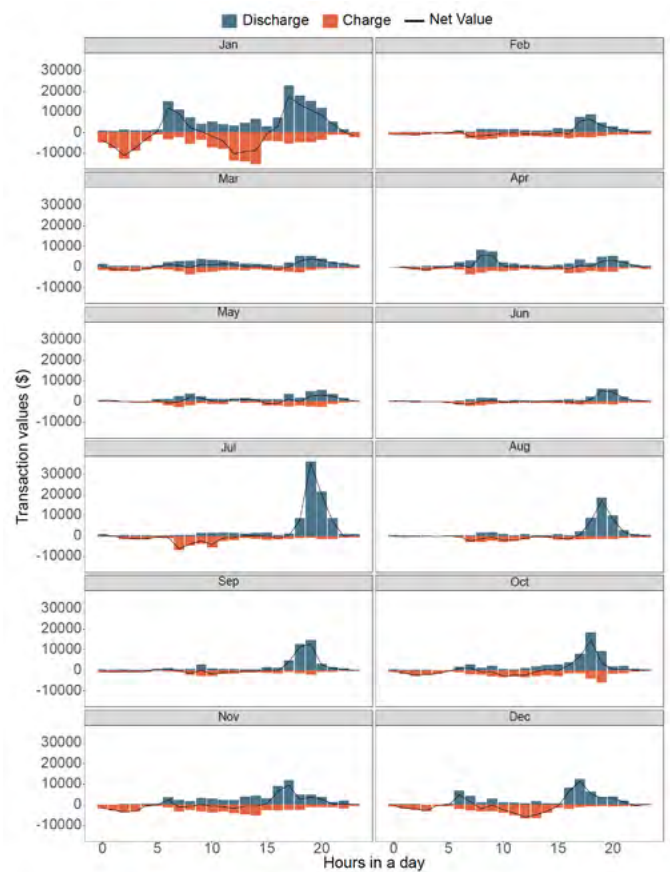
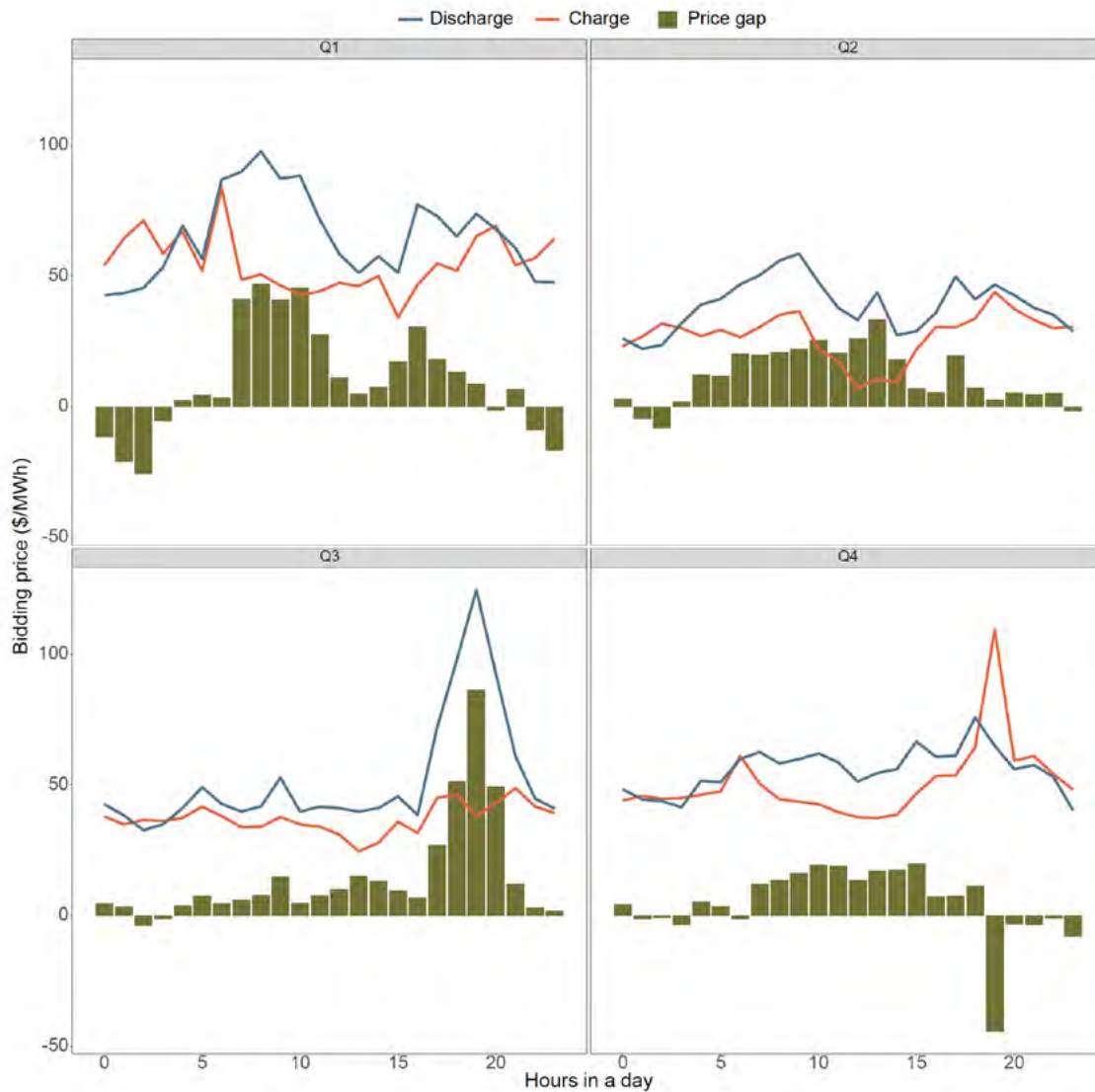


FIGURE 6 (b) Hourly average transaction charges in 2024 by month



Due to its arbitrage-focused strategy, Moss Landing storage project generates a disproportionate share of revenues in a limited number of hours. As shown in Figure 7, 75% of its energy and ancillary market revenues in 2024 was earned in just 15% of trading hours. Primarily, these high-value trading hours occurred between 17:00 - 20:00 during January, July, August and September. This concentration highlights the project's ability to target and capitalize the high-value intervals, generating outsized returns during periods of significant grid stress.

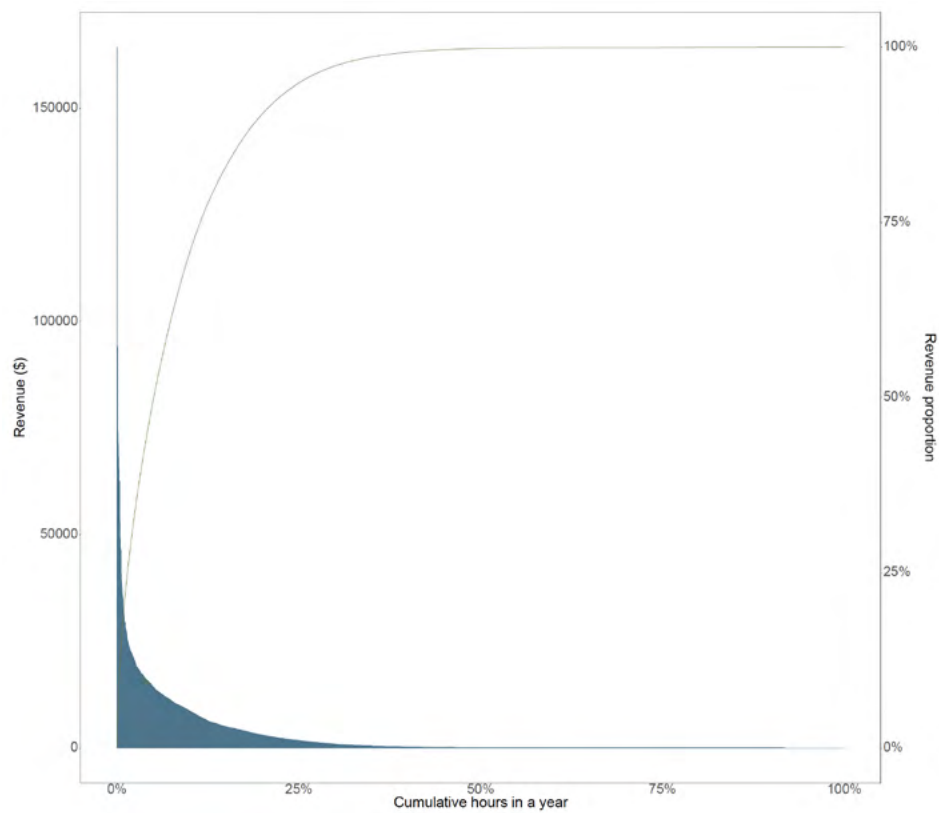
FIGURE 7 Hourly average bidding prices and arbitrage spreads in 2024 (by quarter)



Hence, the operational pattern of the Moss Landing storage project closely tracks shifts in demand and is strongly driven by price signals in the energy market. The underlying logic is simple but effective: buy low in off-peak hours, when solar generation is abundant, and sell high during peak hours. This strategy allows Moss Landing to maximize its load-shifting ability and capture value from California's wholesale market volatility.

In summary, capacity payment serves as the financial backbone for Moss Landing storage project, supported by long-term contracts under California's RA program. These contracts ensure revenue certainty and are reinforced by cost-sharing via CAM. Moreover, while the energy and ancillary service markets are more volatile, they provide opportunities for strategic revenue optimization, particularly during periods of high demand or grid stress. By responding to price volatility and leveraging its flexibility, Moss Landing captures significant value through energy arbitrage.

FIGURE 8 Hourly revenue distribution in energy market across the year



Simply put, this case study illustrates a broader revenue model for battery storage: secure stable income through capacity payments, and capture additional value from high-value intervals in energy and ancillary service markets. Importantly, such a model depends on two key conditions: a competitive wholesale market that accurately reflects the value of battery storage as a flexible resource and a well-developed regulatory framework that supports long-term capacity payment.

CONCLUSION AND RECOMMENDATIONS

California's practical experience shows that energy storage can capture substantial value during periods of high demand and grid stress, with its flexibility rewarded accordingly. Adjustments to spot market rules, including easing price limits, can allow prices to more accurately reflect real-time grid conditions. Greater price volatility, particularly during peak hours on high-demand days, would incentive participation in energy and ancillary service markets and improve profitability for storage investors.

From a policymaker's perspective, further development of spot markets is essential to unleash the full market potential of battery storage and reflect its true value as a flexible resource. As summarized in Section 2, Chinese provinces currently adopt three dominant revenue models for energy storage projects, yet none fully captures the flexibility and system value that battery storage can offer. Across all three models, compensation is either based on fixed pricing or spot market bidding with limited price volatility - conditions that offer insufficient arbitrage opportunities to generate sustainable revenues. Price caps in spot markets were relatively low, in which scarcity of flexible resources, including battery storage, is not adequately compensated, especially during periods of high demand or low renewable output. Moreover, under Model (2) and Model (3), ancillary services are often narrow in scope or compensated at fixed prices, constraining potential revenue streams. To enhance the economic viability of battery storage, ancillary services markets should be expanded to allow a broader range of services at more market-reflective prices, and energy pricing structures should be further reformed to better align with real-time system conditions and more accurately reflect the market value of energy.

Capacity payments are a foundational revenue stream for energy storage projects. Rather than serving as a supplementary income, they can become a primary revenue source. Currently, most provincial pilots rely on one- to three-year payments. Provinces should be encouraged to refine and expand their capacity compensation mechanisms. Notably, the 2024 release of new rules for electricity market operation by the NDRC emphasized the marketization of capacity payments, introducing competition among coal, storage, and other resources.⁶⁹ Although this shift will likely occur gradually, it signals a meaningful change in policy direction.⁷⁰ In line with this shift, moving toward longer-term capacity contracts is critical to ensure stable and predictable returns and strengthen investor confidence in energy storage investments. Ultimately, favorable regulatory frameworks, such as a more formal resource adequacy mechanism, will be necessary to support such long-term capacity payment mechanisms.

⁶⁹ NDRC (2024).

⁷⁰ Williams, S (2024).

REFERENCES

1. Anhui Department of Industry and Information Technology. (2024, June 27). Reply Letter Regarding Proposal No. 1492 of the Second Session of the 14th Provincial People's Congress. <https://m.bjx.com.cn/mnews/20240627/1385663.shtml>
2. Anhui Power Trading Center. (2024, December 3). Basic Operating Rules for Anhui Power Spot Market (Settlement Trial Operation, Version 5) – Draft for Comments. <https://m.bjx.com.cn/mnews/20241203/1414477.shtml>
3. BJX. (2019, May 21). 2019 Energy Storage White Paper Released: China's Cumulative Operational Electrochemical Energy Storage Capacity to Reach 1.9 GW by Year-End. <https://m.bjx.com.cn/mnews/20190521/981719.shtml>;
4. BJX. (2020, May 20). 2020 Energy Storage Industry Research White Paper. <https://mguangfu.bjx.com.cn/mnews/20200520/1074202.shtml>;
5. BJX. (2023, March 10). Revision of Shandong's "Two Rules": New Energy Projects That Fail to Install Required Energy Storage Will Be Shut Down at Twice the Corresponding Capacity. <https://guangfu.bjx.com.cn/news/20230310/1293944.shtml>;
6. BJX. (2025, January 23). NEA: New Energy Storage Installed Capacity to Exceed 70 Million Kilowatts in 2024. <https://m.bjx.com.cn/mnews/20250123/1424444.shtml>
7. California Energy Commission. (n.d.-a). Clean Energy and Pollution Reduction Act—SB 350. <https://www.energy.ca.gov/rules-and-regulations/energy-suppliers-reporting/clean-energy-and-pollution-reduction-act-sb-350>
8. California Energy Commission. (n.d.-b). SB 100 Joint Agency Report. <https://www.energy.ca.gov/sb100>
9. California ISO (2023). Special Report on Battery Storage. <https://www.aiso.com/Documents/2022-Special-Report-on-Battery-Storage-Jul-7-2023.pdf>
10. California ISO (n.d.). Resource Adequacy: The need for sufficient energy supplies. <https://www.aiso.com/documents/resource-adequacy-fact-sheet.pdf>
11. California Public Utilities Commission. (2018). Resolution E-4949. <https://docs.cpuc.ca.gov/publisheddocs/published/g000/m240/k050/240050937.pdf>
12. CESA (2024, June 18). California state assembly passes energy storage bill AB 2514—Landmark energy storage legislation now heads to senate. <https://storagealliance.org/news/state-assembly-passes-energy-storage-bill>

13. China Electricity Council. (2025 March). 2024 Annual Statistical Data on Electrochemical Energy Storage Power Stations. https://www.sohu.com/a/877032049_289755 ; BJX. (2025, January 23). NEA: New Energy Storage Installed Capacity to Exceed 70 Million Kilowatts in 2024. <https://m.bjx.com.cn/mnews/20250123/1424444.shtml>
14. China Energy News. (2023, July 3). From Encouragement to Penalty for Non-Compliance: Storage Allocation Ratios and Duration Requirements Are on the Rise Mandatory Energy Storage for New Energy Projects Is Becoming Increasingly Stringent. https://paper.people.com.cn/zgnyb/html/2023-07/03/content_26004692.htm
15. East China Energy Regulatory Bureau. (2024, September 3). Implementation Rules for Ancillary Services Management in the East China Regional Power Market. <https://m.bjx.com.cn/mnews/20240903/1398476.shtml>
16. Electric Investor Owned Utility Areas. (n.d.). Retrieved May 12, 2025, from <https://gis.data.ca.gov/documents/CAEnergy:electric-investor-owned-utility-areas/about>
17. Electric Quarterly Report. (n.d.). <https://www.ferc.gov/power-sales-and-markets/electric-quarterly-reports-eqr>
18. Federal Energy Regulatory Commission. (n.d. -a). Understanding and Participating in California ISO (CAISO) Processes. <https://www.ferc.gov/understanding-and-participating-california-iso-caiso-processes>
19. Federal Energy Regulatory Commission. (n.d. -b). Order No. 841. <https://www.ferc.gov/media/order-no-841>
20. Gansu Department of Industry and Information Technology. (2024, July 1). Gansu Power Spot Market Management Implementation Rules (Provisional Version V3.1 for Settlement Trial Operation). <https://news.bjx.com.cn/html/20240701/1386260.shtml>
21. Gansu Regulatory Office of NEA. Provisional Rules for the Operation of Gansu Province's Power Ancillary Services Market. <https://gsb.nea.gov.cn/Upload/2022921174615.pdf>
22. General Office of Henan Government. (2023, June 30). Implementation Opinions on Accelerating the Development of New-Type Energy Storage. <https://fgw.henan.gov.cn/2023/06-30/2770724.html>
23. Guangdong DEVELOPMENT AND REFORM COMMISSION & Guangdong Energy Bureau. (2023, May 30). Several Measures to Promote the Development of New Energy Storage Stations. https://drc.gd.gov.cn/ywtz/content/post_4192578.html
24. Guangdong DEVELOPMENT AND REFORM COMMISSION. (2023, June 7). Notice on Matters Related to the Allocation of Energy Storage for New Energy Power Generation Projects. https://drc.gd.gov.cn/ywtz/content/post_4194913.html
25. Guangdong DEVELOPMENT AND REFORM COMMISSION. (2024, October 9). Notice on the Trial Electricity Tariff Compensation Mechanism for Independent Energy Storage Stations and Related Matters (Draft for Comments). <https://drc.gd.gov.cn/hdjlpt/yjzj/answer/39497>

26. Guangdong Energy Bureau & Southern Regulatory Bureau of the NEA. (2023, April 4). Implementation Plan for the Participation of New Energy Storage in the Power Market of Guangdong Province. https://drc.gd.gov.cn/snyj/tzgg/content/post_4148213.html
27. Guangdong Power Exchange Center. 2024 Guangdong Power Market Annual Report. <https://m.bjx.com.cn/mnews/20250303/1429714.shtml>
28. Guangxi DEVELOPMENT AND REFORM COMMISSION. (2023, April 5). Several Measures for Accelerating the Development of New Energy Storage Demonstration Projects in Guangxi (Trial). <https://www.escn.com.cn/news/show-1571746.html>
29. Hebei DEVELOPMENT AND REFORM COMMISSION. (2025, January 1). Notice on Matters Related to Improving the Pilot Electricity Pricing Policy for Independent Energy Storage. <https://m.bjx.com.cn/mnews/20250327/1434273.shtml>
30. Hebei Power Trading Center. (2024, February 26). Southern Hebei Power Spot Market Series Rules Version 3.0. <https://news.bjx.com.cn/html/20250226/1429034.shtml>
31. Hunan DEVELOPMENT AND REFORM COMMISSION. (2024, October 8). Hunan Province Power Spot Market Trading Implementation Rules (Trial Operation Version 3.0) (Draft for Comments). <https://m.bjx.com.cn/mnews/20241008/1403682.shtml>
32. Hunan Regulatory Office of NEA. (2023, May 22). Hunan Province Power Ancillary Services Market Trading Rules (2023 Edition). https://hunb.nea.gov.cn/xxgk/zcjd/202309/t20230913_65789.html
33. Hunan Regulatory Office of NEA. (2025, February 21). Supplementary Provisions to the Medium- and Long-Term Power Trading Rules of Hunan Province. https://hunb.nea.gov.cn/dtyw/tzgg/tz/202502/t20250225_277121.html
34. Jiangsu DEVELOPMENT AND REFORM COMMISSION. (2023, July 19). Notice on Issuing Several Measures to Accelerate the High-Quality Development of New Energy Storage. https://fzggw.jiangsu.gov.cn/art/2023/8/3/art_51012_11064875.html
35. Jiangsu Regulatory Office of NEA. (2024, January 3). Jiangsu Power Auxiliary Services (Frequency Regulation) Market Trading Rules (Draft for Comments). <https://www.escn.com.cn/news/show-1640981.html>
36. McNamara, W. (n.d.). Issue Brief—Energy Storage & Resource Adequacy.
37. NDRC & NEA. (2021, August 10). Notice on Encouraging Renewable Energy Power Generation Enterprises to Build or Purchase Load Regulation Resources for Grid Integration. https://www.ndrc.gov.cn/xxgk/zcfb/tz/202108/t20210810_1293396.html
38. NDRC & NEA. (2023, October 12). Notice on Further Accelerating the Development of the Power Spot Market. https://www.gov.cn/zhengce/zhengceku/202311/content_6913560.htm
39. NDRC & NEA. (2025, February 9). Notice on Deepening the Market-Oriented Reform of Feed-in Tariffs for New Energy and Promoting the High-Quality Development of New Energy. https://www.ndrc.gov.cn/xxgk/zcfb/tz/202502/t20250209_1396066.html

40. NDRC. (2024, April 25). Rules for Electricity Market Operation. <https://zfxgk.ndrc.gov.cn/web/iteminfo.jsp?id=20379>
41. NEA. (2023, February 13). As of the end of 2022, the total installed capacity of new-type energy storage projects in operation nationwide reached 8.7 million kilowatts (MW). https://www.nea.gov.cn/2023-02/13/c_1310697043.htm
42. NEA. (2024, January 25). As of the end of 2023, China had completed and put into operation new-type energy storage projects with a total installed capacity of 31.39 million kilowatts (MW) / 66.87 million kilowatt-hours (kWh). https://www.nea.gov.cn/2024-01/25/c_1310761952.htm
43. NEA. (2025, January 23). NEA Holds a Press Conference to Introduce Developments in New Energy Storage. <https://www.nea.gov.cn/20250123/3e5379eb74b94d94bcdaf556268ad466/c.html>
44. Ningxia Development and Reform Commission. (2024a, December 20). Plan for Independent Energy Storage Participation in Medium- and Long-Term Market Transactions in Ningxia. <https://m.bjx.com.cn/mnews/20241220/1418273.shtml>
45. Ningxia Development and Reform Commission. (2024b, November 5). Notice on Promoting the Healthy Development of Energy Storage. https://fzggw.nx.gov.cn/tzgg/202411/t20241105_4717794.html
46. Northeast Regulatory Bureau of the NEA, Inner Mongolia Department of Economy and Information Technology, Inner Mongolia DEVELOPMENT AND REFORM COMMISSION. (2018, August). <https://img.in-en.com/upload/201801/19/17552083231446.pdf>
47. NREL. (2018). <https://docs.nrel.gov/docs/fy20osti/73643.pdf>
48. NREL. (n.d.). Black Start - Grid Modernization. Retrieved May 21, 2025, from <https://www.nrel.gov/grid/black-start>
49. PwC China, 2023 China New Energy Storage Industry Development White Paper. <https://www.pwccn.com/zh/deals/publications/2023-china-new-energy-storage-development-white-paper-jun2023.pdf>
50. RMI. (2024, May 9). 2024 Insights on Power Market Reform and Electricity Pricing System: Ten Key Trends for Market Participants. <https://rmi.org.cn/insights/2024powermarketreviewandoutlook/>
51. Science and Technology Daily. (2025, January 23). China's new energy storage capacity has exceeded 70 million kilowatts (70 GW). https://www.stdaily.com/web/gdxw/2025-01/23/content_290060.html
52. Shandong Energy Bureau. Shandong Province Implementation Rules for the Administration of Power Auxiliary Services. <https://sdb.nea.gov.cn/main/ContentManage/Article/File/2023/07/24/202307241702345061.pdf>
53. Sina Finance. (2025, April 10). The white paper shows that China's new-type energy

- storage market is projected to exceed a cumulative installed capacity of 200 GW. <https://finance.sina.com.cn/chanjing/2025-04-10/doc-inesspws5742102.shtml>
54. South China Energy Regulatory Bureau. (2025, February 19). Implementation Rules for Grid Operation and Ancillary Services Management of New Energy Storage in the Southern Region. <https://m.bjx.com.cn/mnews/20250219/1427811.shtml>
 55. Utility Dive. (n.d.). California looks to next steps as utilities near energy storage targets | Retrieved April 28, 2025, from <https://www.utilitydive.com/news/california-looks-to-next-steps-as-utilities-near-energy-storage-targets/525441/>
 56. Williams, S. (2024, December 18). Changing how coal power plants get paid: What's next for policy in China. Regulatory Assistance Project. <https://www.raponline.org/blog/changing-how-coal-power-plants-get-paid/>
 57. Xinhua. (2025a, January 24). China's new energy storage capacity exceeds 70 million KW. https://english.www.gov.cn/archive/statistics/202501/24/content_WS6793830cc6d0868f4e8ef23f.html; Guiding Opinions on Accelerating the Development of New Energy Storage. (2021, July 23). https://www.ndrc.gov.cn/xxgk/zcfb/ghxwj/202107/t20210723_1291321.html
 58. Xinhua. (2025b, March 24). Inner Mongolia Issues Notice to Accelerate the Development of New Energy Storage. <http://nmg.news.cn/20250327/45bc70e1dd66430aa00136eff9a26a85/c.html>
 59. Xinjiang Development and Reform Commission. (2023, September 1). Officials from the Xinjiang Uygur Autonomous Region Development and Reform Commission Answer Reporters' Questions on the Notice on Further Improving Time-of-Use Electricity Pricing. <https://xjdrcc.xinjiang.gov.cn/xjfgw/c108363/202312/f1dd8b30722e4bec9e66dba5e86e48aa.shtml>
 60. Xinjiang Development and Reform Commission. (2024, May 19). Notice on Establishing and Improving Supporting Policies for the Healthy and Orderly Development of New Energy Storage. <https://xjdrcc.xinjiang.gov.cn/xjfgw/c108361/202309/02a98beb179645fb924a1a52a798bc11.shtml>
 61. Zhejiang DEVELOPMENT AND REFORM COMMISSION & Zhejiang Energy Bureau. (2021, November 3). Implementation Opinions on Accelerating the Demonstration and Application of New Energy Storage in Zhejiang Province. https://fzggw.zj.gov.cn/art/2021/11/9/art_1229123351_2372518.html